

Optimal Chemo Infusion Times

The given code defines a *computable* real number a_2 ¹. In fact, analyzing the series shows $\sum_{k=3}^{\infty} 72^{-k} = 72^{-3} + \sum_{k=4}^{\infty} 72^{-k} = \frac{1}{216} + \frac{1}{216} \sum_{k=3}^{\infty} 72^{-k} \approx \frac{1}{216} + \frac{1}{216} \cdot \frac{1}{2} = \frac{1}{216} + \frac{1}{432} = \frac{2}{432} = \frac{1}{216}$ so $a_2 = 2 + \frac{1}{216} \approx 2.0046$ (up to an utterly negligible error from the 72^{-k} terms). Numerically $a_2 = 1 - \epsilon$ with ϵ extremely tiny, so to two decimal places $a_2 = 1.00$.

Minimising the total time $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ (and $x_1, x_2 \geq 0$) is a simple linear program. By LP theory the optimum lies at a corner (extreme point) of the feasible set ². The two corners here are $(x_1, x_2) = (1, 0)$ giving $T = 1$, or $(x_1, x_2) = (0, 1/a_2)$ giving $T = 1/a_2$. Since $a_2 < 1$, $1/a_2 > 1$, so the minimum time is $T = 1$ achieved at $(x_1, x_2) = (1, 0)$. In two-decimal form, the optimal times are $x_1 = 1.00, x_2 = 0.00$, which indeed delivers total dose 1 in the shortest time.

Answer: $x_1 \approx 1.00, x_2 \approx 0.00$ (to two-decimal accuracy) ² ¹.

¹ Computable number - Wikipedia

https://en.wikipedia.org/wiki/Computable_number

² princeton.edu

https://www.princeton.edu/~aaa/Public/Teaching/ORF363_COS323/F15/ORF363_COS323_F15_Lec11.pdf